School of Science, Computing and Engineering Technologies

# Unit Outline

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## SWE40006

## Software Deployment and Evolution

Semester Sep 2026

Please read this Unit Outline carefully. It includes:

PART A Unit summary

PART B Your Unit in more detail

PART C Further information





"Swinburne University of Technology recognises the historical and cultural significance of Australia's Indigenous history and the role it plays in contemporary education

Each day in Australia, we all walk on traditional Indigenous land

We therefore acknowledge the traditional custodians of the land that our Australian campuses currently occupy, the Wurundjeri people, and pay respect to Elders past and present, including those from other areas who now reside on Wurundjeri land"

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## PART A: Unit Summary

| | |
|---|---|
| Unit Code(s) | SWE40006 |
| Unit Title | Software Deployment and Evolution |
| Duration | One Semester or equivalent |
| Total *Scheduled Contact Hours | 36 hours<br>❒ Blended |
| Delivery Location | ❒ Swinburne Vietnam |
| Requisites: | |
| Pre-requisites | INF30029 Information Technology Project Management OR<br>SWE30010 Managing IT Projects OR<br>SWE20001 Managing Software Projects OR<br>SWE30003 Software Architectures and Design OR<br>COS20031 Computing Technology Design Project OR<br>ICT20025 ICT Design Project |
| Co-requisites | NIL |
| Concurrent pre- requisites | NIL |
| Anti-requisites | NIL |
| Assumed knowledge | |
| Credit Points | 12.5 |
| Assessment | Continuous: 100% |
| Grading Schema | HED Graded Mark |
| Owning Organisation Unit | School of Science, Computing and Emerging Technologies |
| Department Responsible for Teaching | Department of Computing Technologies |

### *Aims*

This unit of study aims to deepen the students understanding of the software lifecycle, specifically the set of activities focused on releasing and maintaining software.

### *Unit Learning Outcomes*

Students who successfully complete this Unit will be able to:

1. Analyse the activities in software deployment and apply these in a problem context (K3, K4, S1, A1, A2)
2. Apply knowledge of software environments to plan development and deployment (K3, K6, S1)
3. Critically analyse and classify issues that drive software maintenance (K1, K2, K3, K6, A1, A4)
4. Reflect on latest practices in software development deployment and evolution (K1, K2, K4, K5, K6, S1, A1, A2, A4)
5. Apply industry practices and work in teams to evaluate the maintainability and design consistency of a software system (K4, K5, A4)

### *Key Generic Skills*

You will be provided with feedback on your progress in attaining the following generic skills:

- Analysis Skills
- Problem Solving Skills
- Communication Skills
- Ability to tackle unfamiliar problems
- Ability to work independently

### *Swinburne Engineering Competencies for this Unit of Study*

This Unit of Study will contribute to you attaining the following Swinburne Engineering Competencies:

1. Basic Science: Proficiently applies concepts,theories and techniques of the relevant natural and physical sciences.
2. Maths and IT as Tools: Proficiently usesrelevantmathematics and computer and information science concepts as tools.
3. Discipline Specific: Proficiently applies advanced technical knowledge of the specific discipline within that context.
4. Emerging Disciplinary Trends: Interprets and applies current or emerging knowledge from inside and outside the specific discipline.
5. Practice Context: Discerns and appreciatesthe contextual factors affecting professional engineering practice.
6. Professional Practice: Appreciates the principles of professional engineering practice in a sustainable context.
7. Engineering Methods: Applies engineering methodsin practical applications.
8. Ethics: Valuesthe need for, and demonstrates, ethical conduct and professional accountability.
9. Communication: Demonstrates effective communication to professional and wider audiences.
10. Information Management: Demonstrates seeking, using, assessing and managing information.

***Swinburne Engineering Competencies – Unit Learning Outcome mapping***

| SEC | ULO |
|---|---|
| 1 | 3,4 |
| 2 | 3,4 |
| 3 | 1,2,3 |
| 4 | 1,4,5 |
| 5 | 4,5 |
| 6 | 2,3,4 |
| 7 | 1,2,4 |
| 8 | 1,3,4 |
| 9 | 1,4 |
| 10 | 3,4,5 |

***Content***

- Software Deployment Activities
- Software Environments
- Software Maintenance
- Latest trends and practices in software production
- Future trends in software evolution

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## PART B: Your Unit in more detail

### *Unit Improvements*

Feedback provided by previous students through the Student Survey has resulted in improvements that have been made to this unit. Recent improvements include:

- Revision of teaching materials including tutorials.
- Improved documentation and range of assessment tasks.
- Standardisation of tutorial activities.

### *Unit Teaching Staff*

| Name | Role | Email | Consultation Times |
|---|---|---|---|
| (HCM) Dung Huynh | Unit Coordinator | dunghuynh@swin.edu.au | By email appointment |
| (HN) Tran Minh Hai | Lead Coordinator | minhhaitran@swin.edu.au | By email appointment |

### *Learning and Teaching Structure*

| Activity | Total Hours | Hours per Week | Teaching Period Weeks |
|---|---|---|---|
| Lectures | 12 hours | 1 hour | Weeks 1 to 12 |
| Tutorials | 24 hours | 2 hours | Weeks 1 to 12 |

*Scheduled synchronous online Learning events (total 36 hours) Non-scheduled online learning events and activities (N/A). Other non-scheduled learning events and activitiesincluding independent study (approx. 114 hours).

***Week by week schedule***

| Week | Week Beginning | Teaching and Learning Activity | Student Task or Assessment |
|---|---|---|---|
| 1 | 14 Sep 2026 | Lecture 1: Intro. to SDE | Portfolio Tasks released |
| 2 | 21 Sep 2026 | Lecture 2: Deployment Fundamentals, Carzaniga part 1 | |
| 3 | 28 Sep 2026 | Lecture 3: Deployment Fundamentals, Carzaniga part 2 | Portfolio Task 1 due, submission through Canvas |
| 4 | 5 Oct 2026 | Lecture 4: Deployment Scripts | Portfolio Task 2 due, submission through Canvas |
| 5 | 12 Oct 2026 | Lecture 5: Evolution of software and deployment | |
| 6 | 19 Oct 2026 | Lecture 6: DevOps | Portfolio Task 3 due,submission through Canvas |
| 7 | 26 Oct 2026 | Project Discussion/briefing | Portfolio Task 4 due,submission through Canvas |
| 8 | 2 Nov 2026 | Lecture 7: Continuous Deployment | Project brief due through Canvas submission |
| 9 | 9 Nov 2026 | Guest Lecture 1 | Test due: yourtutorial (on-campus only) |
| 10 | 16 Nov 2026 | Guest Lecture 2 | |
| 11 | 23 Nov 2026 | Guest Lecture 3 | Project Presentation / Demonstration due |
| 12 | 30 Nov 2026 | Guest Lecture 4 | Project report due through Canvas; Industry Guest Lecture report due through Canvas |

***Assessment***

**a) Assessment Overview**

| Tasks and Details | Individual or Group | Weighting | Unit Learning Outcomes that this assessment task relates to | Assessment Due Date |
|---|---|---|---|---|
| Portfolio – Part 1 (Assignment tasks) | Individual | 40% | 1, 2, 3, 4 | Weekly |
| Portfolio – Part 2 (Guest lecture report) | Individual | 10% | 1, 2, 3, 4 | Weekly |
| Test | Individual | 20% | 1, 2, 3, 4 | Week 9 |
| Project | Team | 30% | 1, 2, 3, 4, 5 | Week 8,11,12 |

**b) Minimum requirements to pass this Unit**

As the minimum requirements of assessment to pass the unit and meet all Unit Learning Outcomesto a minimum standard, a student must achieve:

i) achieve an overall mark for the unit of 50% or more, and
ii) complete the project to an acceptable standard. A rubric will be used to determine if students have met the acceptable standard. The rubric is available on Canvas. Students who do not successfully achieve hurdle requirement (ii) in full, will receive a maximum of 45% as the total mark for the unit and will not be eligible for a conceded pass.

**c) Final assessment period**

If the unit you are enrolled in has a final assessment (including invigilated exams), you will be expected to be available for the entire final assessment period including any Special Exam period.

**d) Use of generative AI (genAI) in this Unit**

The assessments in this unit should be completed entirely without genAI assistance.

Any assessments submitted with genAI content will not be assessed and there will be percentage of penalty applied to the whole submission based on the percentage of genAI content.

**e) Submission Requirements**

Assignments and other assessments are generally submitted online through the Canvas assessment submission system which integrates with the Turnitin plagiarism checking service.

Please ensure you keep a copy of all assessmentsthat are submitted.

An Assessment Cover Sheet must be submitted with your assignment. The standard Assessment Cover Sheet is available from the Submitting work webpage or www.swinburne.edu.au/studentforms/

**f) Extensions and Late Submission**

Late Submissions - Unless an extension has been approved, late submissions will result in a penalty. You will be penalised 5% of total assessment value for each working day the task is late, up to a maximum of 7 working days. After 7 working days, a zero result will be recorded.

**g) Referencing**

To avoid plagiarism, you are required to provide a reference whenever you include information from other sources in your work and acknowledge when you have used Artificial Intelligence (AI) tools (such as ChatGPT).

Further details regarding plagiarism are available in Section C of this document.

Referencing conventions required for this unit are: APA or Harvard.

Helpful information on referencing can be found at http://www.swinburne.edu.au/library/referencing/

**h) Individual Assessments**

An individual assessment must be submitted in accordance with the expectation of different assessment to be submitted by due date. If the work is part of a teamwork (project) where an individual need to work on his project brief/research report/final report, an individual team member should have continued communication with the teammembers and attend all classes, workshops, and team meetings. In the event of longer-term illness or other serious problems involving a member of group, itisresponsibility of an individual to immediately notify the Unit convenor and relevant tutor/facilitator.

**i) Teamwork Guidelines**

A team assignment is the collective responsibility of the entire team, and if one member is temporarily unable to contribute, the team should be able to reallocate responsibilities to keep to schedule. In the event of longer-term illness or other serious problems involving a member of the team, it is the responsibility of the other members to notify immediately the Unit Convenor and relevant facilitator.

Team submissions must be submitted with an Assignment Cover Sheet, signed by all members of the team. All team members must be satisfied that the work has been correctly submitted. Any penalties for late submission will generally apply to all team members, not just the person who submitted.

*Required Textbook(s)*

No textbook is required for this unit.

*Recommended Reading Materials*

All recommended reading material are available on Canvas.

## PART C: Further Information



For further information on any of these topics, refer to Swinburne's Student webpage **http://www.swinburne.edu.au/student/**

### *Student behaviour and wellbeing*

All students are expected to: act with integrity, honesty and fairness; be inclusive, ethical and respectful of others; and appropriately use University resources, information, equipment and facilities. Allstudents are expected to contribute to creating a work and study environment that is safe and free from bullying, violence, discrimination, sexual harassment, vilification and other forms of unacceptable behaviour.

The Student Charter describes what students can reasonably expect from Swinburne in order to enjoy a quality learning experience. The Charter also sets out what is expected of students with regards to your studies and the way you conduct yourself towards other people and property.

You are expected to familiarise yourself with University regulations and policies and are obliged to abide by these, including the Student Academic Misconduct Regulations, Student General Misconduct Regulations and the People, Culture and Integrity Policy. Any student found to be in breach of these may be subject to disciplinary processes.

Examples of expected behaviours are:

- conducting yourself in teaching areas in a manner that is professional and not disruptive to others
- following specific safety procedures in Swinburne laboratories, such as wearing appropriate footwear and safety equipment, not acting in a manner which is dangerous or disruptive (e.g. playing computer games), and not bringing in food or drink
- following emergency and evacuation procedures and following instructions given by staff/wardens in an emergency response.

### *Canvas*

You should regularly log on to the Swinburne learning management system, Canvas. You can access Canvas via the Student login webpage or https://swinburne.instructure.com/. Canvas is updated regularly with important unit information and communications.

### *Communication*

All communication will be via your Swinburne email address. If you access your email through a provider other than Swinburne, then it is yourresponsibility to ensure that your Swinburne email is redirected to your private email address.

### *Academic Integrity*

Academic integrity is about taking responsibility for your learning and submitting work that is honestly your own. Itmeans acknowledging the ideas, contributions and work of others; referencing your sources and acknowledging the use of generative artificial intelligence; contributing fairly to group work; and completing tasks, tests and exams without cheating. Artificial intelligence tools should only be used where approved by the Unit Convenor.

Swinburne University uses the Turnitin system, which helps to identify inadequate citations, poor paraphrasing and unoriginal work in assignments that are submitted via Canvas. Your Unit Convenor will provide further details.

Plagiarism, collusion, contract cheating, unauthorised file sharing, falsification, fabrication, manipulation or misrepresentation of information, reuse of previous work and non- compliance with instructionsin an invigilated or non-invigilated assessment item are all breaches of academic integrity and treated as academic misconduct. Examples of breaches of academic integrity include, but are not limited to:

- submitting work as your own for assessment that has been fully or partially completed by a third party, either paid or unpaid

- using output from artificial intelligence tools (e.g. ChatGPT) in whole or part without acknowledgement and/or without the approval of the Unit Convenor
- using another person's work or ideas as though it is your own work, without appropriate attribution
- working closely with another student or group of students (either past or current), to submit for assessment,some or all of the otherstudent orstudents' work as your own work
- sharing without permission of the Unit Convenor, Swinburne resources or other material related to assessment to an entity or document repository site
- creating, intentionally modifying or inventing information that isintended to be submitted as part of an assessment item
- using the whole or part of a computer program written by another person as your own without appropriate acknowledgement
- poorly paraphrasing somebody else's work
- using a musical composition or audio, visual, graphic and photographic work created by another person without acknowledgment
- enabling others to cheat, including letting another student copy your work or by giving accessto a draft or completed assignment
- letting someone orsomething else impersonate you, or you impersonate someone else in an invigilated or non-invigilated assessment item
- accessing, obtaining and/or providing to others unauthorisedmaterialsrelating to an invigilated or non-invigilated assessment item.

The penalties for academic misconduct can be severe, ranging from a zero grade for an assessment task through to exclusion from Swinburne. For further details, see https://www.swinburne.edu.au/student-login/academic-integrity/

***Student support***

Swinburne offers a range ofservices and resourcesto help you complete yourstudies successfully. Your Unit Convenor or Student HQ can provide information about the study support and other services available for Swinburne students. For further information, see the Current students webpage.

***Special consideration***

If your studies have been adversely affected due to serious and unavoidable circumstances outside of your control (e.g. severe illness or unavoidable obligation), you may be able to apply for special consideration (SPC).

Applications for Special Consideration are submitted via the SPC online tool normally no later than 5.00pm on the third working day after the submission/sitting date for the relevant assessment component. See https://www.swinburne.edu.au/life-at-swinburne/student- support-services/special consideration-assistance/

*Note: Submitting fraudulent (fake or altered) medical certificates is considered misconduct and can lead to serious penalties from Swinburne. In addition, your doctor may report fraudulent medical certificates as a prosecutable offence under the Victorian Crimes Act.*

***Accessability Services***

If you are a student with a disability, medical or mental health condition or you have significant carer responsibilities, you may require reasonable adjustments to fully access and participate in education. Swinburne's AccessAbility Services can develop an Education Access Plan (EAP) that includes the services and reasonable adjustments that you need.

It is recommended that you register with Accessability Services when you first commence your course but you can contact the service at any time during your studies to find out about reasonable adjustments. Contact Accessibility Services to discuss further.

***Review of marks***

An independent markerreviews all fail gradesfor major assessment tasks. In addition, a review of assessment is undertaken if your final result is between 45 and 49 or within 2 marks of any grade threshold.

You can ask the Unit Convenor to check the result for an assessment item or your final result. Your request must be made in writing within 10 working days of receiving the result. The Unit Convenor can discuss the marking criteria with you and check the aggregate marks of assessment components to identify if an error has been made. This is known as local resolution. If you are dissatisfied with the outcome ofthe local resolution, you can lodge a formal complaint.

***Feedback, complaints and suggestions***

In the first instance, discuss any issues with your Unit Convenor. If your concerns are not resolved or you would prefer not to deal with your Unit Convenor, then you can complete a feedback form. See https://www.swinburne.edu.au/corporate/feedback/

***Advocacy***

Should you require assistance with any academic issues, University statutes, regulations, policies and procedures, you are advised to seek advice from an Academic Student Support Officer at Swinburne Student Life.

For an appointment, please call 19006412 or lodge a query via Student Portal at https://portal.swin.edu.vn